

River-network information improves validation NSE, but the gain remains 1.00%

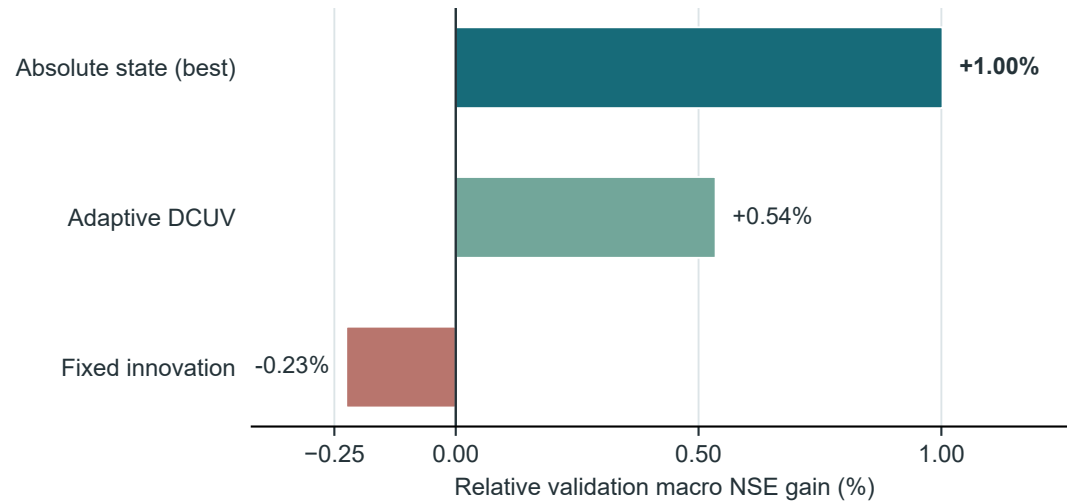
Same-budget no-graph NSE 0.589238 → best directed NSE 0.595144 | 15% target NSE 0.677623

a Controlled attention-value ablation

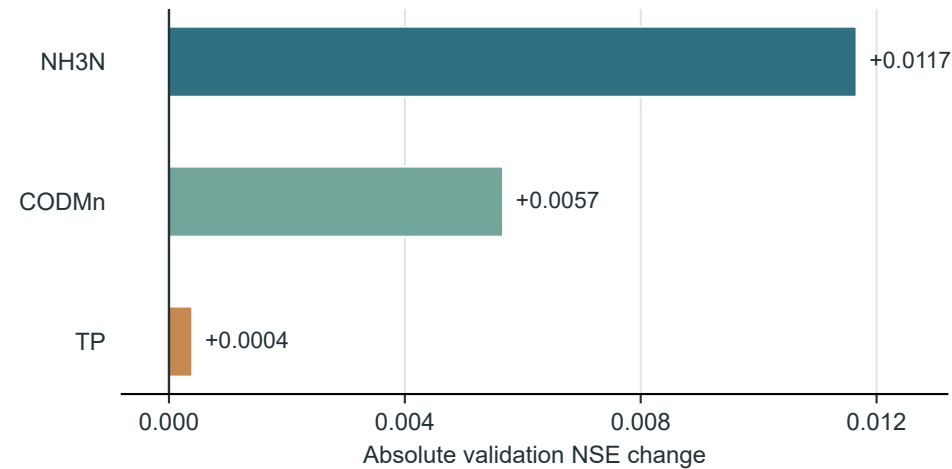
BEST 1.00%

REQUIRED 15%

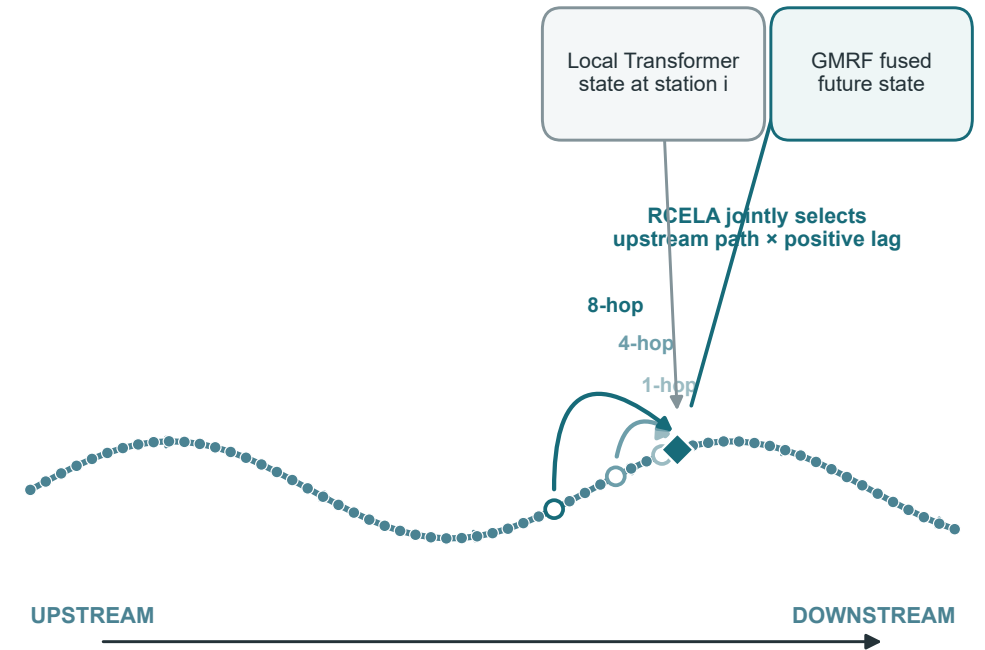
only 6.7% of target



b Best graph model: gain by pollutant



c Directed lag-aware Transformer–GNN recurrence



61 stations | 60 directed edges | 153.0 d cumulative travel-time prior

At forecast day h , attention can read observed history or earlier model states only; current/future target information is never available.